99 — Source Coverage Ledger

99 — Source Coverage Ledger

Revision: 1 **Last modified:** 2026-07-05T14:20:00Z **Status:** Draft — consolidated from `99-source-coverage-ledger.md`.

---

1. Position

This section is the §11.4.118 proof-of-completeness for the MVP documentation consolidation. It maps every canonical source document in `docs/research/mvp/final/` to the implementation section that carries its distinctive content, and it surfaces any source ideas that are not yet fully reflected.

The authoritative version lives at `../../99-source-coverage-ledger.md`; this README is the navigable consolidation.

---

2. Source → implementation mapping

Source	Implementation section(s)	Distinctive contribution
`SPECIFICATION.md`	00 — Executive Summary, 02 — System Architecture	Architectural spine, roles, principles, roadmap, decision register
`00-product-scope-and-principles.md`	01 — Product Scope	Product definition, personas, scope, parity matrix, decisions D1–D8
`01-data-plane.md`	03 — Data Plane	`Transport` trait, transports, routing, multihop, MTU, DAITA

Source	Implementation section(s)	Distinctive contribution
`02-control-plane.md`	04 — Control Plane, 08 — API Contracts	Go monolith, data model, events, coordinator, API, reconciliation
`03-client-core-and-ui.md`	05 — Client Core & UI	FFI surface, Flutter UI, platform shims, state management
`04-security-privacy-pki.md`	06 — Security, Privacy & PKI	Zero-trust, identity, PKI, no-logging, kill-switch, PQ, threat model
`05-repo-layout-tooling-and-helix-ecosystem.md`	07 — Infrastructure & DevOps	Repo layout, codegen, `helixvpnctl`, substrates, ecosystem
`06-phase0-spike-wbs.md`	11 — Guides & FAQs	Phase-0 gates G1–G6, milestones S0–S8
`07-phase1-mvp-wbs.md`	11 — Guides & FAQs	Phase-1 MVP WBS, 8-criteria DoD, SLOs
`08-phase2-parity-wbs.md`	11 — Guides & FAQs	Phase-2 parity WBS, P2-AC + SLOs
`09-phase3-reach-wbs.md`	11 — Guides & FAQs	Phase-3 extended-reach WBS, gates G20–G26
`10-testing-acceptance-and-qa.md`	09 — Testing & QA	§11.4.169 test types, evidence model, acceptance gates
`11-deep-research-appendix.md`	12 — Appendix: Research	Consolidated cited research (10 angles)
`99-source-coverage-ledger.md`	this section	Source-coverage proof and residual gaps
`MASTER_INDEX.md`	00 — Executive Summary	Full document tree, statuses, quality items
`REFINEMENT_NOTES.md`	00 — Executive Summary	Pass-1→pass-N punch-list

Nano-detail volumes

Volume	Dir	Implementation section
V0 Meta	`v00-meta/`	00, this section
V1 Product	`v01-product/`	01, 08
V2 Data Plane	`v02-data-plane/`	03
V3 Control Plane	`v03-control-plane/`	04, 08
V4 Clients	`v04-client/`	05
V5 Security	`v05-security/`	06
V6 Deploy	`v06-deploy/`	07
V7 Execution	`v07-execution/`	11
V8 Testing	`v08-testing/`	09
V9 Research	`v09-research/`	12
V10 Design	`v10-design/`	10

External design-system index

File	Implementation section
`docs/design/README.md`	10 — Design System

---

3. Residual coverage gaps

The following distinctive source ideas are deliberately not adopted or are recorded as open.

G1 — `[03_ZAI]` disaster-recovery completeness — RESOLVED

`v06-deploy/disaster-recovery.md` now consolidates the RTO/RPO budget, KMS-encrypted backups, and Terraform-driven region-failover runbook that the original ledger called for. Independently re-verified 2026-07-04. No further action.

G2 — `[06_GRK]` sing-box transport framework — deliberate divergence

Proposed: use sing-box as the universal transport multiplexer. **Decision: not adopted**. The protocols it carries (Hysteria2, Shadowsocks, UDP-over-TCP) are reflected in the Phase-2 transport set, but the framework itself conflicts with the Rust-core decision and the iOS NE memory ceiling. The custom Rust `helix-transport` crate gives byte-for-byte client↔edge reuse.

Rationale recorded: `99-source-coverage-ledger.md` §G2 and `03 — Data Plane.

G3 — `[09_GCT]` Fyne desktop UI — deliberate divergence

Proposed: use Fyne for desktop/web client. **Decision: not adopted**. Flutter reaches all 8 required platforms (including Aurora OS and HarmonyOS NEXT) with one codebase and a shared `helix_design` system. Fyne lacks mobile/niche-OS reach.

Rationale recorded: `99-source-coverage-ledger.md` §G3 and `05 — Client Core & UI.

---

4. Decision coverage cross-check

All eight key decisions are surfaced explicitly in `final/`:

Decision	Camp A	Camp B	Surfaced in
D1 obfuscation	MASQUE/QUIC	Hysteria2 + Salamander	`01-data-plane.md`, `08-phase2-parity-wbs.md`, `11-deep-research-appendix.md`
D2 client core	Rust	Go	`03-client-core-and-ui.md`, `11-deep-research-appendix.md`
D3 event bus	Redis → NATS	NATS from start	`02-control-plane.md`, `08-phase2-parity-wbs.md`
D4 subnet collision	ULA/4via6	CGNAT	`02-control-plane.md`, `08-phase2-parity-wbs.md`
D5 edge language	Rust	Go	`01-data-plane.md`, `06-phase0-spike-wbs.md`
D6 transport topology	single protocol	asymmetric per-leg	`01-data-plane.md`, `11-deep-research-appendix.md`
D7 ambition	lean	Connectivity-OS	

Decision	Camp A	Camp B	Surfaced in
			`00-product-scope-and-principles.md`, `SPECIFICATION.md`
D8 licensing	source-available + commercial	pure OSS	`SPECIFICATION.md` §9

---

5. Cross-references

- Authoritative ledger → `../../99-source-coverage-ledger.md`
- Synthesis evidence base → `../../v09-research/_SYNTHESIS.md`
- Gap analysis → `../../../../../reviews/mvp-final/findings/phase1-docs-gap-analysis.md`

Source: `docs/research/mvp/final/99-source-coverage-ledger.md`.